Final Project: Sentiment Analysis of a Product or Service

Start Assignment

- Due Aug 2 by 11:59pm
- Points 100
- Submitting a file upload

Final Project: Sentiment Analysis of a Product or Service

Instructions

Group (2-3 members) or Individual Work

For your final project, you will apply everything you have learned to build and evaluate a sentiment analysis model for a real-world product or service of your choice. You will be a data scientist solving a business problem: understanding customer feedback.

Objectives

•Independently manage an end-to-end machine learning project from problem definition to conclusion.
•Select and justify an appropriate public dataset for your chosen topic.
•Apply text preprocessing and feature extraction techniques (e.g., TF-IDF) suitable for NLP.
•Train and evaluate at least two different classification models for sentiment analysis.
•Analyze model performance using appropriate metrics and interpret the results in the context of your business problem.
•Communicate your findings and the story of your data through a well-documented Jupyter Notebook.

Project Ideas & Dataset Suggestions

Choosing a good topic and dataset is the first step. Your goal is to find customer reviews that are labeled as positive, negative, or neutral. Here are some ideas and places to find datasets. All of these are well-suited for Google Colab.

Project Ideas (Beginner-Friendly):

•Movie Reviews: Are reviews for a specific movie genre (e.g., horror, comedy) generally positive or negative?
•Restaurant Reviews: What is the sentiment of Yelp or Zomato reviews for a chain of restaurants?
•Product Reviews: Analyze Amazon reviews for a popular product category (e.g., headphones, coffee makers).
•Airline Feedback: What is the sentiment of tweets directed at a specific airline?
•Hotel Reviews: Analyze TripAdvisor reviews for hotels in a specific city.

Suggested Public Datasets:

•**Kaggle: A great resource for finding clean, public datasets.**

•**[IMDb Dataset of 50K Movie Reviews](https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews) (https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews)** : A classic dataset, perfect for binary (positive/negative) sentiment analysis.
•**[Amazon Fine Food Reviews](https://www.kaggle.com/snap/amazon-fine-food-reviews) (https://www.kaggle.com/snap/amazon-fine-food-reviews)** : A large dataset of food reviews, includes a 1-5 star rating you can convert to sentiment.
•**[Twitter US Airline Sentiment](https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment) (https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment)** : Tweets about major US airlines, labeled as positive, negative, or neutral.
•Hugging Face Datasets: A library with thousands of easily accessible datasets.
•You can load datasets like imdb or yelp_review_full directly in Colab with a few lines of code.

Step-by-Step Instructions

1.Choose Your Team, Topic, and Dataset
•Decide if you will work individually or in a group of 2-3.
•Select a project topic and find a suitable public dataset. Your dataset must have text data (the reviews) and a label (e.g., positive/negative, 1-5 stars).
2.Use the Starter Notebook
•A file named Final_Project_Starter_Notebook.ipynb is provided.
•Upload it to Google Colab. This notebook provides a structured framework with markdown cells for guidance and empty code cells for you to fill in.
3.Complete the Project Notebook (Parts 1-6)
•Part 1: Project Definition: State your business problem and describe your dataset.
•Part 2: EDA: Explore your text data. What are the most common words? How long are the reviews? Is your dataset balanced?
•Part 3: Preprocessing & Feature Extraction: Clean the text data (e.g., remove punctuation, stopwords) and convert it into numerical features using TfidfVectorizer.
•Part 4: Modeling: Train at least two models (e.g., LogisticRegression, NaiveBayes, RandomForestClassifier).
•Part 5: Evaluation: Evaluate your models using accuracy, precision, recall, and a confusion matrix. Which model is better and why?
•Part 6: Conclusion: Summarize your findings. What is the sentiment of the reviews you analyzed? What are the limitations of your model?
4.Name and Submit Your File
•Rename your notebook to FP_ProjectTopic_YourFullName (e.g., FP_MovieReviews_JaneDoe). If in a group, include all names.
•Submit your single, completed .ipynb file.

Expected Deliverables

•One completed Jupyter Notebook (.ipynb file) containing all your code, outputs, analysis, and interpretations. The notebook should tell a clear and complete story of your project.
Present notebook and explain what you have leanrned

Tips for Success (with Limited Resources)

•Start Small: When exploring, use a sample of your data (df.sample(n=1000)) to speed up initial tests before running on the full dataset.
•Limit Vocabulary: When using TfidfVectorizer, set the max_features parameter (e.g., max_features=5000) to use only the top 5,000 most frequent words. This dramatically reduces memory usage.
•Use Efficient Models: LogisticRegression and MultinomialNB (Naive Bayes) are very fast and memory-efficient, making them excellent choices for text data on Colab.
•Document Everything: Your thought process is as important as your code. Use the markdown cells to explain your decisions.

Self-Check Before Completion

My project topic and dataset are clearly defined.
All 6 parts of the starter notebook are complete.
The notebook runs from start to finish without errors.
My analysis and interpretations are written clearly in the markdown cells.
The final file is named correctly: FP_ProjectTopic_YourFullName.

General Project Guidelines

Group & Individual Work Policy
This project can be completed individually or in groups of up to 3 members. If working in a group, all members are expected to contribute equally. The final submission should be a single notebook with all group members' names on it.
File Naming Convention
•Final Project Notebook: FP_ProjectTopic_YourFullName.ipynb
•If in a group: FP_ProjectTopic_Name1_Name2.ipynb
•Examples:
•FP_AirlineTweets_MariaGarcia.ipynb
•FP_AmazonReviews_AlexChen_BenC

Rubric for grading:

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Needs Improvement (D/F)	Points
1. Problem Definition & Dataset Selection	Clearly defines a meaningful business problem and explains why the dataset is	Business problem and dataset are explained adequately.	Basic explanation provided but lacks clarity or justification.	Problem definition unclear or dataset inappropriate/missing explanation.	10

	appropriate. Dataset is relevant, clean, and well-described.				
2. Exploratory Data Analysis (EDA)	Thorough EDA with meaningful visualizations and insights (class balance, review length, common words, etc.). Strong interpretation of findings.	Good EDA with some visualizations and explanations.	Limited EDA or minimal interpretation.	Very little/no EDA or incorrect analysis.	15
3. Text Preprocessing & Feature Engineering	Text cleaning is comprehensive and well-justified. TF-IDF or similar technique correctly implemented and explained.	Appropriate preprocessing completed with minor issues.	Basic preprocessing completed but lacks explanation or depth.	Preprocessing incomplete, incorrect, or missing.	15
4. Model Development	At least two appropriate models trained correctly with thoughtful comparison and parameter choices.	Two models implemented correctly with basic comparison.	Models implemented with limited explanation or errors.	Only one model used or implementation incorrect/incomplete.	20
5. Model Evaluation & Metrics	Uses multiple evaluation metrics correctly (accuracy, precision, recall, confusion matrix). Strong interpretation of results and comparison between models.	Evaluation metrics used correctly with adequate interpretation.	Limited evaluation or interpretation.	Evaluation missing, incorrect, or poorly explained.	20
6. Conclusions & Business Insights	Clear conclusions tied back to the business problem. Discusses limitations and future	Conclusions are reasonable and somewhat connected to project goals.	Basic summary provided with limited insight.	Weak or missing conclusion.	10

	improvements thoughtfully.				
7. Notebook Organization & Documentation	Notebook is exceptionally organized, easy to follow, well-commented, and professionally presented. Markdown explanations are clear and complete.	Notebook is organized and mostly clear.	Notebook somewhat difficult to follow or lacks sufficient documentation.	Poor organization, missing explanations, or difficult to read.	5
8. Presentation & Explanation of Learning	Student/group explains workflow, findings, and lessons learned clearly and confidently. Demonstrates strong understanding.	Explanation demonstrates good understanding with minor gaps.	Basic explanation with limited depth.	Unable to explain project decisions or learning outcomes clearly.	5